and tools developed by global AI majors outside of India.

Bridging this gap are efforts such as Bhashini of MeiTY (ministry of electronics and IT), apart from collaborative projects between academia and industry such as the AI for Bharat initiative that has since been spun off as a start-up by the name Sarvam.AI. It is noteworthy that several independent efforts have also emerged—from Zoho to Ola—with indigenously developed language models for AI. While much of the mainstream media attention has been focused on language models, much thanks to the hype generated by ChatGPT over the past year, there are several specialised start-ups that have been focused on niche areas of AI with a uniquely Indian perspective. One example is Paralaxiom, a start-up founded by a group of IIT Bombay alumni, which focuses on developing computer vision-based AI models for real-time video analytics.

Start-ups such as Pienomial and Partex, also founded by IIT-B alumni, have been focused on developing AI models and intellectual property to solve problems in life sciences. The acute shortage of trained radiologists has seen an interesting use of AI-based image analysis to assist in diagnosis of X-rays and other medical imaging technologies by start-ups such as the 5C network.

While India may have lagged in the development of large language models (LLMs) that have dominated the public attention on AI, the advances in niche models are consistent with the recent remarks of Infosys chairperson Nandan Nilekani on why it is important for India to prioritise its efforts on specific AI applications.

A primary reason for this is the significant costs associated with high-performance computing infrastructure required to train and operate such large AI models. The accelerated pace of evolution of AI models over the past year owes much to the advances in high-performance computing along with the development of advanced semiconductor chips such as GPUs (graphics processing units) from Nvidia, and TPUs (tensor processing units) from Google, both of which are optimised for AI. The steep costs associated with these semiconductor chips and their limited availability pose a significant barrier to the development of large-scale models in India. While India has been one of the largest sources of engineering talent to global tech majors for both semiconductor design and development of AI, there continues to be a significant gap in indigenous development of these technologies given the huge R&D investments needed. For India to close the gap on these foundational hardware and software technologies necessary for AI development would require the nurturing of an ecosystem of risk-tolerant technology investment at the scale of billions of dollars.

**TAKEAWAYS**

- **Indigenous AI development will need an ecosystem of risk-tolerant technology investment worth billions of dollars**
- **Developing high-performance computing as a 'digital public infrastructure' which start-ups can access is an alternative**
- **Large-scale AI models in India are hampered by the high cost of semiconductors and their limited availability**

An alternative approach proposed by public-private initiatives such as People+ai envisages developing high-performance computing as a digital public infrastructure that can be made available to start-ups and researchers to aid in the training and development of indigenous AI models and applications. Such a sovereign, cloud-based compute infrastructure for India was also advocated by Nvidia CEO Jensen Huang during his recent visit to India and meeting with PM Modi. While collaboration by Nvidia with Indian majors such as Reliance, Yotta and others holds promise, the widespread availability of high-performance computing at a reasonable cost to Indian developers and researchers will likely take time.

Other alternative approaches being attempted include the use of open source for smaller scale models that creatively leverage existing computing infrastructure. India's ability to play at the foundational level in the global arena will however require a leapfrog approach similar to Aadhaar, UPI, etc., that gives India a population-scale advantage. It would require leveraging the power of the Indian market at a scale of a billion users through AI use-cases focused on edge devices such as smartphones and other innovative end-user devices while converging with other edge technologies such as 5G/6G and Direct to Mobile Broadcasting (D2M). India's path to global AI leadership will require both a creative convergence of emerging technologies and public-private collaboration by drawing upon lessons from its success with population-scale Digital Public Goods. ■